

Project Executable Files

Date	10 June 2026
Team ID	xxxxxx
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL (if hosted)	Yes

7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

Voice-Based-Concept-Understanding-Analyser/

```
|
|  └─ uploads/           # Uploaded audio files
|  └─ reports/           # Generated PDF reports
|  └─ temp/              # Temporary waveform images
|  └─ tests/             # Test files
|
|  └─ app.py             # Main Streamlit application
|  └─ model_loader.py    # AI model loading and caching
|  └─ speech_to_text.py  # Whisper transcription module
|  └─ semantic_eval.py   # Sentence-BERT semantic evaluation
|  └─ audio_utils.py     # Audio feature extraction
|  └─ text_utils.py      # Filler word detection
|  └─ scoring_engine.py  # Understanding scoring engine
|  └─ report_generator.py # PDF report generation
|
|  └─ requirements.txt   # Project dependencies
|  └─ packages.txt       # FFmpeg installation (Streamlit Cloud)
|  └─ README.md          # Project documentation
└─ .gitignore           # Git ignore rules
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://voice-based-concept-understanding-analysergh3u9lyi9cnzn8vlpv5.streamlit.app/
Login (Demo) Credentials	Not Required

Platform / Hosting Provider	Streamlit Community Cloud
Repository Link	https://github.com/soumaykohli/Voice-Based-Concept-Understanding-Analyser.git
Demo Video Link	https://drive.google.com/file/d/1DCg-JZ6t1rJo-iNeNwfYsjY4h5oLemOp/view?usp=sharing

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Step 1

Clone the GitHub repository. `git clone https://github.com/your-username/Voice-Based-Concept-UnderstandingAnalyser.git`

Step 2

Navigate to the project directory.

```
cd Voice-Based-Concept-Understanding-Analyser
```

Step 3

Create a virtual environment.

Windows:

```
python -m venv vbcu_env
```

Activate it.

```
vbcu_env\Scripts\activate
```

Step 4

Install all required dependencies.

```
pip install -r requirements.txt
```

Step 5

If running locally on Windows, ensure **FFmpeg** is installed and available in the system PATH.

Step 6

Launch the Streamlit application.

```
streamlit run app.py
```

Step 7

The application will automatically open in your browser.

If it does not, open: <http://localhost:8501>

Step 8

Upload a WAV audio file.

Enter the reference concept.

Click **Analyze Concept Understanding**.

View:

- Speech Transcript
- Semantic Similarity Score
- Audio Features
- Understanding Score
- Waveform Visualization

Download the generated PDF report.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Initial execution may take longer because Whisper and Sentence-BERT models are loaded into memory.	Models are cached using Streamlit caching, reducing processing time for subsequent evaluations.
2	The application currently supports only English-language audio recordings.	Future versions will include multilingual speech recognition and evaluation.
3	The application accepts uploaded audio files only and does not support real-time microphone recording.	Live speech recording will be implemented in future releases.